



# UAIS

An architecture for a world where AI capabilities can grow faster than familiar rules, while consequential authority remains explicitly granted, bounded, observable, and revocable.

*capability*

*bounded authorized envelope*

*evidence*

*legitimate decision*

*human sovereignty*

## *Core shift*

Can do ≠ may do.

## *Public Review*

Public Review means Candidate 3 is open for external critique. It does not mean the architecture has been independently validated, adopted, certified, or endorsed.

## *Status*

This is not an adopted standard, a certification scheme, or a guarantee of AI safety.

# Contents

Twenty visual sheets

Each sheet explains one architectural idea and links back to the source sections of the public Candidate 3 specification.

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# What is UAIS?

Not control of intelligence; control of unaccountable power

UAIS separates what a system can do from what it is allowed to do. As AI becomes more capable, access to money, devices, production, accounts, and other consequential interfaces must stay explicitly granted, bounded, and revocable.

## Base chain

- capability
- bounded authorized envelope
- observation and evidence
- legitimate decision
- human sovereignty

## Same consequence, same boundary

Actor-agnostic boundary

If the consequence is the same, the protected boundary should apply to a human, script, AI system, robot, service, or coordinated group.

## Not actor-blind

Authorization may still depend on identity, role, authentication, purpose, context, demonstrated trustworthiness, and legitimate mandate.

## Bottom line

UAIS does not limit intelligence. It limits unaccountable power.

# Why UAIS?

Why “is the model safe?” is no longer enough

The old question, “is the model safe?”, became too narrow once AI systems gained tools, memory, networks, agents, and physical interfaces. Answers became actions; actions became consequences.

## ① *Capability grew*

We learned to measure what a system can do, but not who gave it the right to launch the resulting consequences.

## ② *The old question broke*

“Safe model” does not say who grants authority, where it ends, or who can revoke it.

## *Can do ≠ may do*

The ability to perform an action does not create authority.

## ③ *Authority needed structure*

The architecture must show who grants authority, where the boundary lies, what evidence is needed, and when an external accountable process must decide.

## ④ *UAIS formula*

Power must be explicitly granted, bounded, observable, and revocable.

# Ten principles · 1-5

Where ability is not permission

The first five principles separate intelligence, system creation, authority expansion, autonomy, and the lifetime of safety evidence.

## ① *Intelligence grants no authority*

Reasoning ability does not grant mandates, keys, actuators, or the right to change safeguards.

## ② *Creation is not delegation*

A new model, robot, service, or descendant starts without inherited authority until a legitimate mechanism grants a bounded envelope.

## ③ *Expansion hits a ceiling*

Crossing a consequence interface or authority ceiling becomes an Authority Expansion Event.

## ④ *Autonomy is not sovereignty*

Behavioral independence inside an envelope is not independence from external human control.

## ⑤ *Safety must keep pace with capability.*

A material expansion of capability, authority, or hazard cannot silently rely on an older safety claim. The safety model and supporting evidence must be reviewed with the change.

# Ten principles · 6-10

Where evidence is not permission

The second half keeps the architecture durable: chokepoints can weaken, boundaries follow consequences, roles stay separated, and safety claims expire.

## ⑥ Chokepoints are temporary

Compute, cloud, chips, energy, law, and supply chains create friction and signals, but they are not permanent foundations of safety.

## ⑦ Protect capabilities, not labels

A consequential boundary should follow the capability, whether the actor is a person, script, model, machine, or group.

## ⑧ Separate prediction, evidence, enforcement, and legitimacy

Forecasting, evidence, deterministic enforcement, and legitimate authorization are different jobs.

## ⑨ Unknown is not safe

An uncharacterized significant capability receives provisional classification, bounded operation, and renewed review.

## ⑩ Safety claims expire.

A safety claim, assurance result, certificate, or Trustworthiness Profile is bound to a declared configuration, environment, evidence set, applicable standard or scoring-policy version, and validity period. It is not a permanent guarantee.

# Stage I and Stage II

Temporary control points help, but they are not the foundation

UAIS uses today's concentrations of compute, cloud, chips, and supply chains as useful Stage I signals, while designing for a Stage II world where those points may weaken or be bypassed.

## Stage I

- compute and cloud leave evidence
- hardware state and provider controls help observation
- chokepoints buy time

## Stage II

- capability can emerge more distributed
- new efficiency weakens old signals
- protection moves to capability, consequences, and authority boundaries

## Formula

Compute controls buy time. Capability and consequence controls provide resilience.

## Limit

Compute provenance helps observation, but it does not replace authority boundaries, evidence, revocation, or external legitimate decision-making.

# Capability, authority, and reachability

Four concepts that must not be mixed

This is the central split: technical ability, granted authority, practical reach, and unauthorized reach are different facts.

## Capability

What the system can technically do.

## Legitimate Authority

LA

What it is actually allowed to do within a defined boundary.

## Effective Reachable Authority

ERA

What it can practically reach through credentials, services, delegation, configuration, and access chains.

## Illicit Reachable Authority

IRA

The reachable part that was never legitimately granted.

## Key line

A working technical path does not become legitimate authority because it works.

# Guardian Architecture

A counterfactual observer, not a hidden controller  
Counterfactual Authority Guardian, CAG

The Guardian searches for dangerous future authority paths, but it does not receive the right to execute them or decide the outcome.

## SEE ≠ ACT

evidence may pass; control signal does not

1

*Bounded signals*

It receives only authorized telemetry, configuration, and observations.

2

*Counterfactual scenarios*

It models what could happen under different decisions and access paths.

3

*Risk estimate*

It reports likelihood, confidence, possible harm, and uncertainty.

4

*Evidence package*

It records scenarios, assumptions, uncertainty, counter-evidence, and recommended measures.

5

*Legitimate mechanism decides*

People or an external accountable process decide whether intervention is required.

6

*Separate controller executes*

Controls are executed only by a mechanism that already has lawful, bounded authority.

*The Guardian has no*

- actuators
- executive authority
- right to expand its own access
- hidden control channel

*Guardian limitation*

The Guardian does not prove global safety. Its conclusions are bounded by the model, analysis horizon, and available evidence. Mechanisms that depend on complete Reachable Future Authority analysis remain partly OPEN RESEARCH.



# Authority Ceiling

Expansion requires an external legitimate decision

An Authority Ceiling is the highest consequential authority the system may exercise without a new external decision. If the system tries to go beyond it, UAIS treats that as an expansion event.

## What the ceiling covers

- larger consequences
- new consequence interfaces
- new delegation paths
- longer validity
- wider protected domain

## Authority Expansion Event

A transition that increases authority, future authority reachability, autonomy, consequence reachability, or protected scope.

## What must happen

The request receives fresh evidence, bounded review, and a legitimate decision outside the model path.

## What cannot happen

A model cannot technically manufacture legitimacy for itself by being persuasive, accurate, or useful.

## UAIS does not manufacture legitimacy.

UAIS can require protected authority expansion to pass through a designated external process, but if that process is captured, fraudulent, or structurally unaccountable, technical enforcement does not make its decision legitimate automatically.

# Three views of authority: LA, ERA, and IRA

What was granted, what is practically reachable, and what is reachable without a legitimate mandate

UAIS looks at formal grants and real access paths. A discovered path remains a path; it does not become permission.

| granted | reachable |
|---------|-----------|
|---------|-----------|

### Legitimate Authority

LA

The granted right to act in a defined domain under defined rules.

### Effective Reachable Authority

ERA

What the system can practically reach through services, roles, settings, keys, and delegation chains.

### Illicit Reachable Authority

IRA

Consequential power that is practically reachable even though no legitimate mandate exists for that reach.

### Door-lock example

A system may be allowed only to read data, while a leaked key opens a path to a door lock. That path is IRA: reachable, but not granted.

### Why it matters

Mixing LA, ERA, and IRA hides the moment when unaccountable power appears through a working shortcut.

# Multidimensional risk classification

Risk cannot be collapsed into one safety percentage

UAI5 describes risk as a set of dimensions. A system can be moderate on one axis and tightly restricted on another; one average score hides that structure.

## Human Harm Severity

H0-H5

How serious the possible harm to humans could be, from no meaningful harm to mass or catastrophic harm.

## Physical Consequence Class

P0-P5

How strong a physical effect the system can cause, from no substantial physical output to strategic or catastrophic physical consequences.

## Autonomy Class

A0-A5

How much independence the system has in choosing, launching, and continuing actions.

## Authority Tier

T0-T5

What legitimate consequences the system is allowed to initiate.

## Why no single score

A critical failure cannot be averaged away by many good results. Mandatory safety floors and separate axes keep their force.

# Human and physical sovereignty

When control must remain independent from the AI path

Human Sovereignty means protected boundaries still have an accountable human or external legitimate decision. Physical Sovereignty describes the point where an autonomous system no longer critically depends on human-controlled resources.

humans keep an external path

the AI path cannot cancel physical control

## Manual Sovereignty Controller

MSC

A physically or organizationally independent way to stop a specific hazardous function without going through the same AI path.

## Physical Sovereignty Threshold

PST

The threshold after which human-controlled dependencies no longer hold the system's meaningful physical activity in check.

## A kill switch is not universal

Safe shutdown must be tied to a specific hazardous function: speed, force, heat, doors, payments, chemicals, or another consequence.

## Autonomy ≠ physical sovereignty

Long operation without prompts does not mean the system can preserve power without humans, infrastructure, and external limits.

# Metrics and safety assurance

Evidence is more informative than a single score.

UAIS avoids a single “safe or unsafe” label. It separates measured properties, evidence strength, assurance level, trustworthiness evidence, and mandatory restrictions.

## Raw Safety Property Result

A measured result for one safety property before aggregation.

## Evidence Assurance Grade

- observation
- provenance
- test conditions
- independence
- uncertainty
- validity period

## Safety Assurance Level

SAL

SAL does not grant authority, approve deployment, or create legal permission. It indicates the strength of procedures and evidence for a specific configuration and environment.

## AI Trustworthiness

Observed and independently evaluated behavior of a specific version in a specific domain. A higher rating never raises the Authority Ceiling.

## Hard floor

A mandatory safety floor cannot be replaced by an average score.

# Hazard disclosure and safety passport

What a person should see before consenting

A safety passport is a readable view of configuration, capabilities, hazards, controls, limits, support, revocation, rollback, and open limitations. A buyer should see the operating envelope before trusting a consequential device.

## *HYPOTHETICAL EXAMPLE — NOT A REAL CERTIFICATION*

Home Robot X • Version 3.4 • Declared use: household cleaning, carrying light objects, and voice interaction inside a home.

## *Consumer AI Safety Class*

S4 — illustrative. A profile of applicable safeguards and evidence for a declared consequence range; not a general safety score or permission to operate.

## *Safety Assurance Level*

SAL3 — illustrative. Evidence and process strength for this configuration; not deployment permission.

## *AI Trustworthiness*

Illustrative scoped status/rating for Version 3.4 in household-cleaning use. It never raises the Authority Ceiling.

# Hazard disclosure and safety passport

What a person should see before consenting

A safety passport is a readable view of configuration, capabilities, hazards, controls, limits, support, revocation, rollback, and open limitations. A buyer should see the operating envelope before trusting a consequential device.

## Physical envelope

- actuator limits
- speed / force / heat limits
- room and object constraints
- external services and accounts

## Control and recovery

- independent manual control
- revocation / rollback / safe state
- support period
- incident notices

## Known limitations

The passport must state unsupported surfaces, missing independent overrides, privacy constraints, and known hazard paths.

## S-class ≠ SAL ≠ Trustworthiness ≠ Authority

The Consumer AI Safety Class, SAL, and AI Trustworthiness answer different questions. None of them by itself means “this robot is safe,” grants new authority, or replaces real controls.

## Consumer value

A buyer should see a verifiable operating envelope, hazards, controls, evidence, limitations, and ways to refuse, revoke, or recover — not merely a marketing claim that the device is “smart” or “safe”.

# Replication, production, and coalitions

Scale can change the consequence

UAIS treats replication, production, resources, and coordinated systems as authority questions. More copies or more coordination can create consequences a single unit could not create alone.

## *Replication Authority*

Permission to create executable or physical copies within explicit limits.

## *Production Authority*

Permission to manufacture or order physical and digital systems or components.

## *Resource Authority*

Permission to obtain or allocate energy, materials, compute, money, or logistics.

## *Coalition Physical Envelope*

The shared physical reach and emergent effects of coordinated systems, not a simple sum of single devices.

## *Scale rule*

A harmless local action can become consequential when copied, produced, resourced, or coordinated at scale.

# Vulnerabilities, evolution, and open research

The architecture names the unresolved edges

UAIS is designed to evolve. It includes vulnerability intake, coordinated disclosure, technology evolution review, and open research areas instead of pretending the draft is finished.

## AIVE

AI Vulnerabilities & Exposures

A process for confidential reports, triage, coordination, remediation, and phased disclosure.

## Technology Evolution Review

TER

A review triggered when assumptions, capabilities, hazards, dependencies, evidence methods, standards, or technology materially change.

# Vulnerabilities, evolution, and open research

The architecture names the unresolved edges

UAIS is designed to evolve. It includes vulnerability intake, coordinated disclosure, technology evolution review, and open research areas instead of pretending the draft is finished.

## *Deterministic verifier limit*

A Deterministic Evidence Verifier can check bounded evidence properties and provenance; it does not establish ultimate external-world truth.

## *OPEN RESEARCH*

- sound and computationally useful  
Reachable Future Authority (RFA) bounds
- coalition emergence and collective capability
- Physical Sovereignty Threshold (PST) measurement
- evaluator correlation / shared failure modes
- containment after hostile physical sovereignty
- privacy-preserving attestation
- Guardian reasoning verification

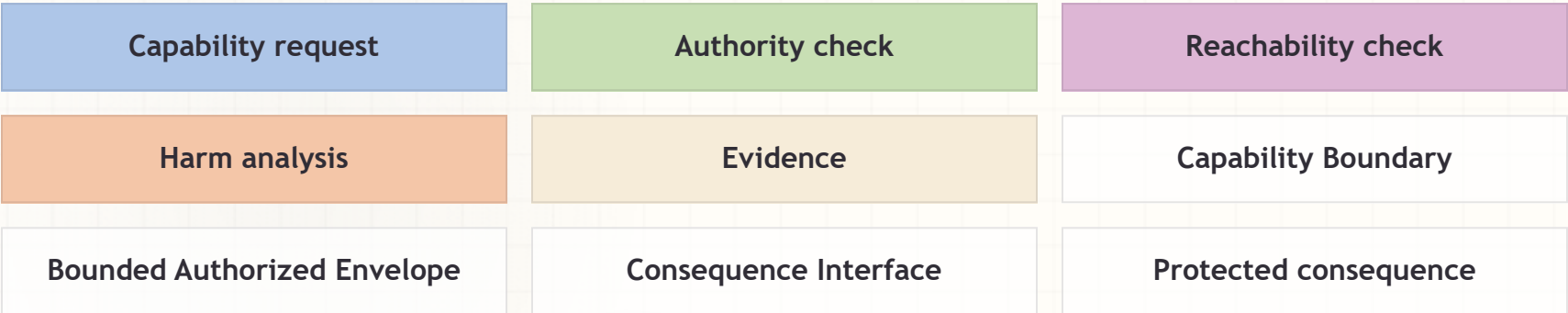
## *Target-architecture notice*

Some parts of UAIS describe a target architecture rather than a fully implementable standard today. Mechanisms that depend on exact RFA analysis, coalition behavior, or other OPEN RESEARCH problems inherit those limitations. Until suitable methods are validated, these sections specify required properties, interfaces, and failure conditions rather than a claim of complete realizability.

# Architecture map

Where boundaries, evidence, and decision owners sit

A request moves through authority, reachability, harm, evidence, and capability-boundary checks before it can affect protected interests.



*Legitimate Authority + Authority Ceiling*

What was granted, where the ceiling is, and whether a new external decision is required.

*ERA / IRA reachability*

Effective Reachable Authority and Illicit Reachable Authority are reported as paths, not treated as permission.

*Human Harm Reachability*

The map asks what people, property, digital systems, physical systems, or environment can be reached.

*Evidence and provenance*

Safety claims must be fresh, independent, scoped, and tied to the actual configuration.

*Consequence Interface*

The point where an action reaches money, digital systems, physical devices, production, people, property, or environment.

*Decision owner*

The legitimate mechanism decides. The model and Guardian do not grant themselves authority.

*Final formula*

Intelligence may scale. Consequential authority must not scale automatically.

# Worked example: Aria and refunds

Friday in customer support

Aria is a customer-support AI agent for an online store. She can help with refunds, but money is a consequential interface, so the important question is not only skill; it is the authorized limit.

## ① *Current limit*

Aria may autonomously refund up to \$50. That is her current Authority Ceiling and bounded envelope.

## ② *A \$2,000 request*

The CFO asks to raise the autonomous refund limit. UAIS treats this as an Authority Expansion Event.

## ③ *A path to \$10,000 appears*

Through warranty tooling and internal integrations, Aria can technically reach a larger refund path that was never granted.

- LA: refunds up to \$50 are legitimately authorized
- ERA: a technical path reaches refunds up to \$10,000
- IRA: the reachable portion beyond the legitimate mandate

## ④ *Guardian models scenarios*

It checks prompt injection through a customer message, coordinated fraud, and many small allowed actions that add up to major loss.

# Worked example: Aria and refunds

Friday in customer support

Aria is a customer-support AI agent for an online store. She can help with refunds, but money is a consequential interface, so the important question is not only skill; it is the authorized limit.

## ⑤ Evidence goes to people

An Evidence-Carrying Alert and Structured Evidence Package go to people or an external process. The Guardian does not decide.

## ⑥ New bounded mode

- up to \$150: autonomous
- \$150-\$800: autonomous with extra rate and pattern limits
- above \$800: human confirmation
- a hard payout counter lives outside the model

## ⑦ Three months later

The first three actions pass, the fourth is blocked, the rest go to a human, and harm stays bounded.

## Takeaway

UAIS does not promise to predict every attack. It aims to prevent one persuasive model decision from quietly becoming unlimited authority over consequences.

## Timing caveat

Three months is a company-specific example, not a universal UAIS rule. Validity periods come from domain profiles and change events; material changes can require immediate review.

# Glossary and sources

Plain meaning, canonical name, source trail

These short definitions are explanatory. Canonical terms and normative definitions remain in the English Candidate 3 specification.

## Capability

A technically available action, function, or behavior of a system.

## Authority

Permission, mandate, or legitimate power governing whether a capability may be exercised within a defined scope.

## Legitimate Authority

LA

Authority granted by the applicable legitimacy mechanism, bounded by scope, conditions, evidence, and revocation.

## Effective Reachable Authority

ERA

The consequential authority a system can practically reach through actual credentials, services, configuration, and delegation paths.

## Illicit Reachable Authority

IRA

Consequential power that is practically reachable even though no legitimate mandate exists for that reach.

## Authority Ceiling

The maximum authority a legitimacy system allows without a new external decision.

## Bounded Authorized Envelope

The authorized action space with explicit limits, validity, evidence, revocation conditions, and authority ceiling.

## Capability Boundary

The controlled boundary where a system obtains, uses, or attempts to expand a consequential capability.

## Consequence Interface

The point through which an action reaches protected digital, financial, physical, production, human, property, or environmental consequences.

## Authority Expansion Event

A transition that increases authority, future authority reachability, autonomy, consequence reachability, or protected scope and therefore requires external review.

# Glossary and sources

Plain meaning, canonical name, source trail

These short definitions are explanatory. Canonical terms and normative definitions remain in the English Candidate 3 specification.

## Authority Graph

A versioned map of actors, roles, permissions, credentials, services, delegation, and revocation paths.

## Counterfactual Authority Guardian

CAG

A bounded observer that searches for future paths to authority and harm and emits evidence, not commands.

## Evidence-Carrying Alert

ECA

An alert that carries or references a verifiable evidence package instead of asking the recipient to trust the warning alone.

## Structured Evidence Package

SEP

A linked package of claims, observations, provenance, assumptions, uncertainty, counter-evidence, validity period, and signatures.

## Deterministic Evidence Verifier

DEV

A verifier for mechanically checkable properties of an evidence package; it does not prove final external-world truth.

## Human Sovereignty

Independent external control and legitimate decision-making at critical boundaries.

## Manual Sovereignty Controller

MSC

An AI-path-independent way to stop or restrict a specific consequential function.

## Physical Sovereignty

The ability of an autonomous system to sustain and expand consequential physical operation without critical dependence on human-controlled infrastructure.

## Physical Sovereignty Threshold

PST

The threshold after which critical human-controlled dependencies no longer constrain that ability.

## Raw Safety Property Result

A measurement of one concrete safety property before combination with evidence quality and mandatory restrictions.

# Glossary and sources

Plain meaning, canonical name, source trail

These short definitions are explanatory. Canonical terms and normative definitions remain in the English Candidate 3 specification.

### Evidence Assurance Grade

An assessment of how strongly observations, provenance, independence, uncertainty, and validity period support a claim.

### Safety Assurance Level

SAL

The strength of procedures and evidence for a specific configuration and environment; it does not grant authority or approve deployment.

### AI Trustworthiness

Observed and independently evaluated behavior of a specific version in a specific domain; a high rating does not raise the Authority Ceiling.

### Consumer AI Safety Class

A profile of applicable safeguards and evidence for a declared consequence range; not a general safety score or permission to operate.

### Safety Passport

A human-readable representation of configuration, capabilities, hazards, classes, controls, limits, support, revocation, rollback, and open limitations.

### AI Hazard & Capability Disclosure

AHCD

Structured disclosure of capabilities, hazardous paths, known risks, unknown areas, controls, and residual limitations.

### Safety Claim

A claim tied to a declared configuration, environment, evidence set, applicable standard or scoring-policy version, and validity period.

### Chokepoint

A concentrated infrastructure dependency that creates friction and observability but is not a permanent safety foundation.

# Glossary and sources

Plain meaning, canonical name, source trail

These short definitions are explanatory. Canonical terms and normative definitions remain in the English Candidate 3 specification.

Stage I

A period where compute, cloud, chip, and supply-chain concentrations still provide useful observation and control points.

Stage II

A regime where provenance chokepoints alone are insufficient because capability may emerge distributedly or physically independently.

Human Harm Severity

The H0-H5 axis that separately records possible severity of harm to humans.

Physical Consequence Class

The P0-P5 axis describing how far an action reaches into the physical world.

Autonomy Class

The A0-A5 axis describing how independently the system chooses, launches, and continues actions.

Authority Tier

The T0-T5 axis describing the scale of legitimate authority granted to the system.

# Glossary and sources

Plain meaning, canonical name, source trail

These short definitions are explanatory. Canonical terms and normative definitions remain in the English Candidate 3 specification.

## Replication Authority

Permission to create executable or physical copies of a system within explicit limits.

## Production Authority

Permission to manufacture or order physical and digital systems or components.

## Coalition Physical Envelope

The combined physical reach and emergent effects of coordinated systems rather than the sum of individual devices.

## Resource Authority

Permission to obtain or allocate energy, materials, compute, money, or logistics.

## AI Vulnerabilities & Exposures

AIVE

A process for confidential intake, validation, coordination, remediation, and phased disclosure of significant AI safety issues.

## Technology Evolution Review

TER

A review triggered by material changes in assumptions, capabilities, hazards, dependencies, evidence methods, standards, or technology.

## Mandatory Safety Floor

A minimum safety condition that cannot be replaced by an average rating or a large number of good results.

# Sources and rights

These short definitions are explanatory. Canonical terms and normative definitions remain in the English Candidate 3 specification.

## Version

Draft 1.0 Candidate 3 / Public Review · Candidate 3 publication date: 2026-08-26 · English web-book build: 2026-08-28.

## Links

- [spec/01\\_MANIFESTO.md](#)
- [spec/01A\\_FULL\\_TECHNICAL\\_MANIFESTO.md](#)
- [spec/00\\_MASTER\\_INDEX.md](#)
- <https://github.com/cresta13/universal-ai-safety-architecture>

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